

Ch4 Linear Representation Hypothesis: Why Linear?

Source: reading_pack/md/A04_LRH.md | Full textbook: TEXTBOOK_Stability_Identifiability.md

4 LRH

4.1 Park

LRH **c** **f_c**

$$x = \sum c_i \cdot f_i + \text{noise}, \quad c_i = \frac{1}{\sqrt{L}}$$
$$p(\text{next token} \mid x) \propto \exp(u_v \cdot r_L)$$

f_c **c** counterfactual pair **James/Mary**

Log-Odds **softmax CE** **logit** **log-odds** **log-odds**

GD **f** **Park Theorem 1-2**

4.2

Park **linear probe $\approx$ non-linear probe** **steering** **LRH** **superposition**

Park p.1-3, p.4-6 Theorem **Gurnee p.2-4**
